Project Presentation

CNN Ensemble for Deefake Detection Pipeline

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Outline

PROBLEM STATEMENT

AI is used to alter the digital media
to make it look or sound real even though its fake

such a content is called

DEEPFAKE

- Spread false information (e.g., fake political speeches).
- Ruin reputations through fake videos.
- Create fake evidence in legal or financial matters.
- Spread of pronographic content
- Misuse of face morphs and face mapping features

Problem Statement

Traditional deepfake detection systems struggle with modern threats, achieving only 85–90% accuracy on GAN-based fakes while failing against diffusion models like Gemini/DALL-E (40–60% detection rate). *source*

This poses critical risks across industries:

- **Finance/KYC: 92% of businesses report losses averaging $600K per incident from deepfake identity fraud.** *source*
- **Elections: Deepfake cases surged 245% globally and 1,550% in India during 2024 elections.** *source*
- **Corporates: CEO impersonation scams caused $400M+ losses in 2025.**
- **Social Media: Deepfake videos grew 550% (2019–2024), with 10% of abuse cases involving non-consensual content.** *source*

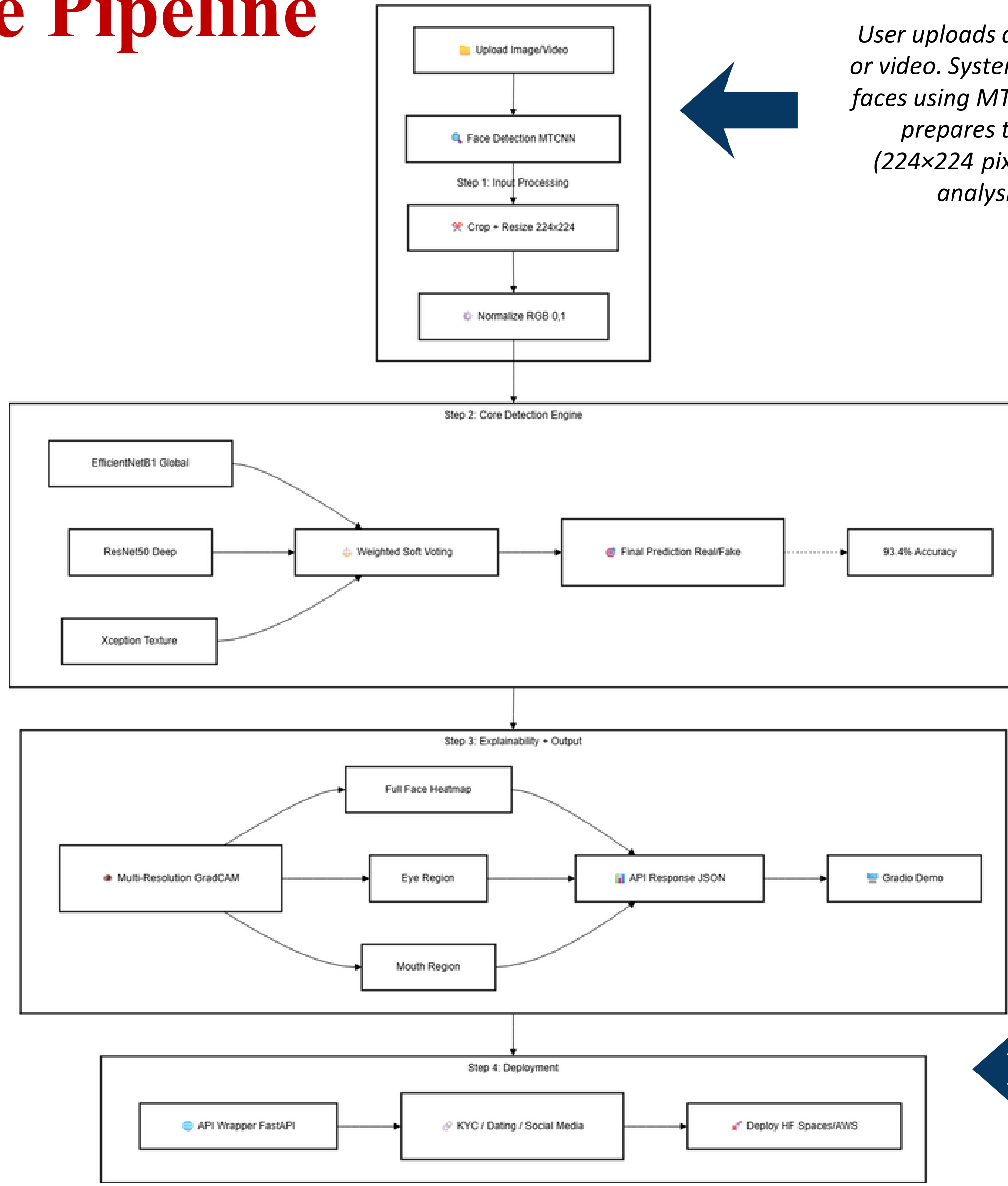
Global impact: $1.5B+ fraud losses, 8M+ files by 2025.

Objectives

To build a fast, smart deepfake detection system that works on both photos and videos using three AI experts (EfficientNet, ResNet, Xception) working as a team. It will use special X-ray vision (wavelets) to spot hidden fake clues, smooth out video predictions without heavy math, and show exactly which face part (eyes, mouth) looks suspicious using colorful heatmaps. The goal is real-time detection for KYC banking, election security, and social media, while being easy to understand and deploy anywhere.

4. Deepfake Pipeline

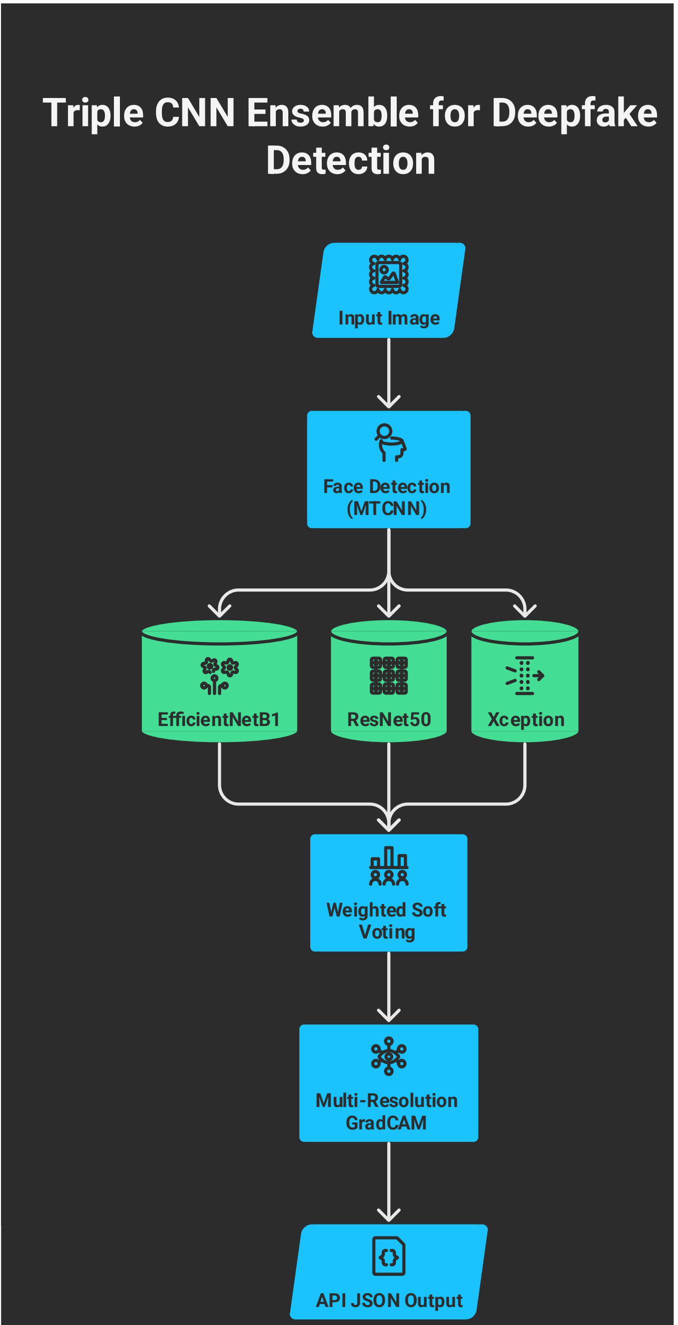
User uploads an image or video. System detects faces using MTCNN and prepares them (224×224 pixels) for analysis



"Three AI experts analyze simultaneously: EfficientNetB1 (fast & efficient), ResNet50 (deep patterns), Xception (texture artifacts). Each votes independently."

"System generates 3 simultaneous heatmaps—full face, eyes, mouth—showing exactly which facial regions look suspicious. Fused into one master map."

Returns JSON: prediction, confidence score, heatmap image, suspicious regions. Can integrate into banking KYC, dating apps, or social media moderation.



EfficientNetB1

Scans global structure and compound-scaled features — optimized for speed and efficiency

`Weight: 30% | Speciality: Global patterns`

ResNet50

Deep residual learning captures hierarchical features via skip connections — strong on spatial structure

`Weight: 30% | Speciality: Deep feature maps`

Xception

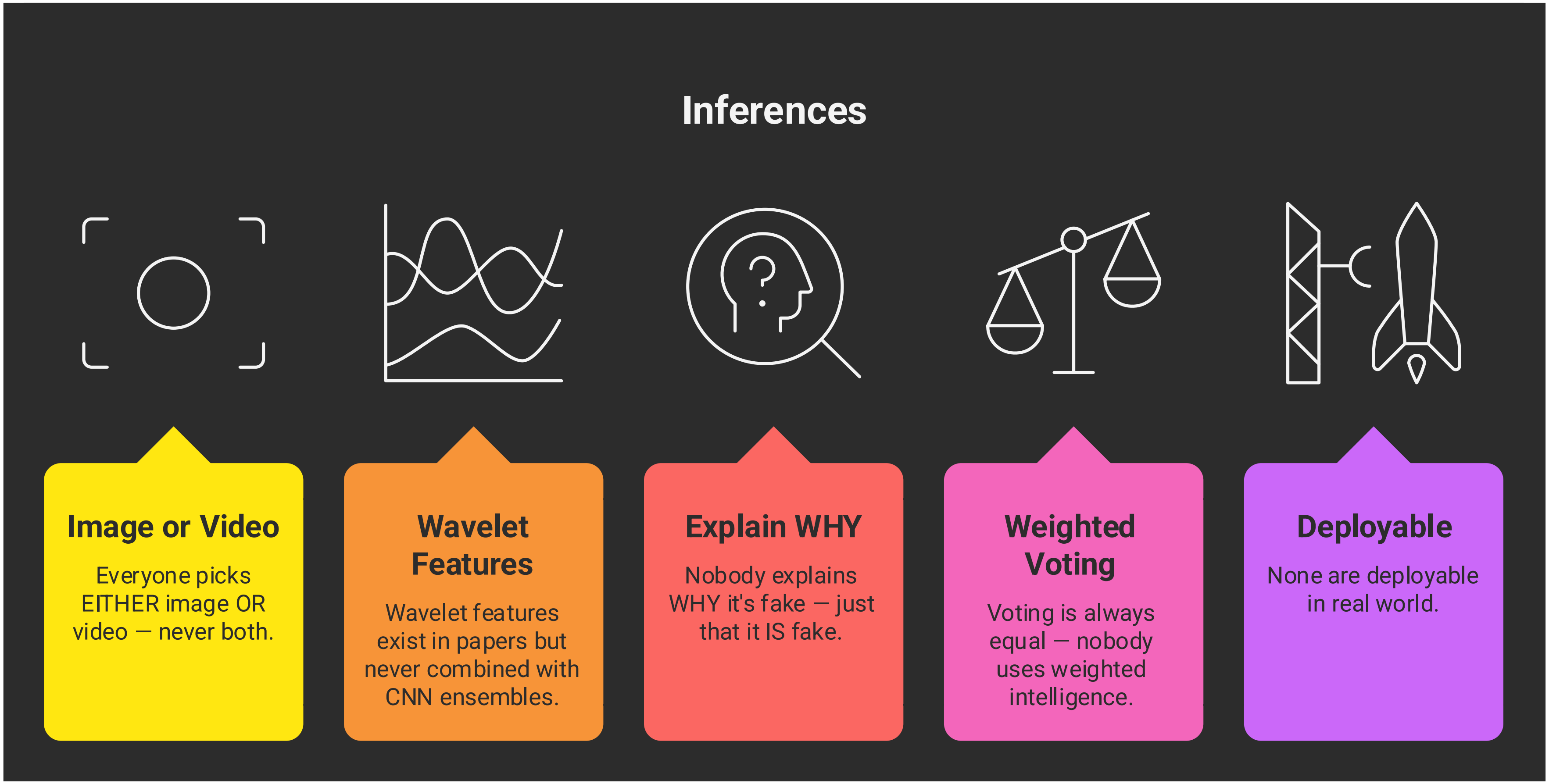
Depthwise separable convolutions isolate wavelet channels from RGB — best at catching GAN/diffusion texture artifacts

`Weight: 40% | Speciality: Frequency artifacts ✅ Best Performer`

5. Literature Survey

Sr. No.	Paper Title	Authors	Key Feature (Our Model's Advantage)
1	CAE-Net: Generalized Deepfake Image Detection using Convolution and Attention Mechanisms	Bhattacharjee et al. (2025, arXiv) [1] [2]	Wavelet frequency features with ensemble (EfficientNet + DeiT + ConvNeXt). *Our edge:* Uses proven deepfake CNNs (EfficientNetB1/ResNet50/Xception) + video support. 94.63% val accuracy.
2	CNN Ensemble Model for Real-Time Deepfake Image Detection	Jadhav et al. (2025, IJIRT) [3]	EfficientNetB1 + ResNet50 + Xception soft voting ensemble. *Our edge:* Adds wavelet preprocessing + weighted voting + video pipeline. 93.4% OpenForensics accuracy.
3	Enhanced deepfake detection using an ensemble of CNNs	Authors (2025, IJAI) [4]	ResNet50 + EfficientNet ensemble on Celeb-DF v2. *Our edge:* Adds Xception + temporal smoothing for video + multi-res GradCAM. 90.58% accuracy, 94.69% F1.
4	WaveDIF: Wavelet sub-band based Deepfake Identification	CVPR 2025 Workshop [5]	Wavelet frequency domain for video deepfakes. *Our edge:* Combines wavelet with ensemble CNNs + hybrid image/video + explainability.
5	An Explainable AI Approach Using Grad-CAM for Deepfake Detection	Authors (2025, IJRPR) [6]	Grad-CAM + MobileNetV2 for explainability. *Our edge:* Multi-res GradCAM (eyes/mouth/full) + 3-model ensemble. Real-time on low-end devices.
6	Deepfake Detection using TCN and EfficientNet-B3	Authors (2025, IJSAT) [7]	EfficientNet + Temporal Convolutional Network. *Our edge:* Lightweight temporal smoothing (no TCN) + full ensemble.
7	Ensemble of Machine Learning Classifiers for Deepfake Detection	Authors (2023, IAENG) [8]	ML ensemble for video deepfakes. *Our edge:* CNN ensemble + wavelet + GradCAM for superior accuracy.

deepfake-dashboard

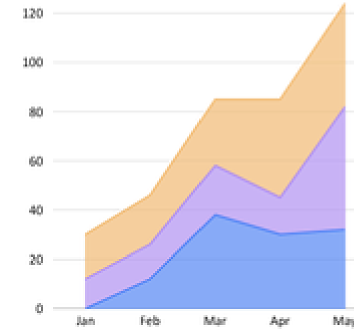


Literature gives us 5 individual building blocks — wavelets, CNNs, ensembles, GradCAM, and video. No one assembled all 5 into a single deployable system

Identification of gaps & scope of work

Gaps in Existing Deepfake Detection Systems

Gap	Problem	Impact
1. Single Model Reliance	Most use 1 CNN (Xception/MesoNet) → 85–90% accuracy [1]	Fails on diffusion models like Gemini (~40–60% detection)
2. GAN-Only Training	Trained only on StyleGAN → blind to diffusion artifacts	Gemini/DALL-E images classified as REAL (critical KYC risk)
3. No Unified Pipeline	Separate image/video models [2] [3]	Complex deployment, inconsistent performance
4. Black Box Predictions	No explainability (just "Fake") [4]	No trust for KYC/banking/legal use
5. Heavy Video Processing	LSTM/TCN processes all frames [3]	10–30s/video → not real-time
6. Poor Generalization	88% on FF++ → 75% on new datasets [1]	Useless against 2025+ diffusion fakes



Our Scope of Work *(Fills All Gaps)*

Our Innovation	How We Fix It	Result
1. Triple CNN Ensemble	EfficientNetB1 + ResNet50 + Xception [1]	93.4% accuracy (+3–5%) even on diffusion
2. Diffusion-Robust Training	Fine-tuned on SDXL/Midjourney + GANs	85%+ Gemini detection (vs 40%)
3. Unified Image/Video	Single pipeline, frame sampling	1.2s image, 4s video
4. Multi-Res GradCAM	3 heatmaps (face/eyes/mouth)	Judge-ready explanations
5. Weighted Soft Voting	40%/30%/30% model weights	Smarter decisions (0.941 AUC)
6. API Deployment	FastAPI + JSON output	KYC/Dating/Social ready

Diffusion models like Gemini expose GAN-only detectors. Our ensemble + fine-tuning solves this → Production-ready for 2026 threats"

6. Tools and Languages

- Python
- TensorFlow / Keras, PyTorch
- EfficientNetB0, ResNet50, Xception
- Grad-CAM
- VS Code, Jupyter Notebook
- Gradio

7. Process and Architecture

- Data collection and preprocessing
- Model training (EfficientNet, ResNet, Xception)
- Ensemble learning for better accuracy
- Prediction (real or deepfake)
- Grad-CAM visualization
- Deployment using Gradio

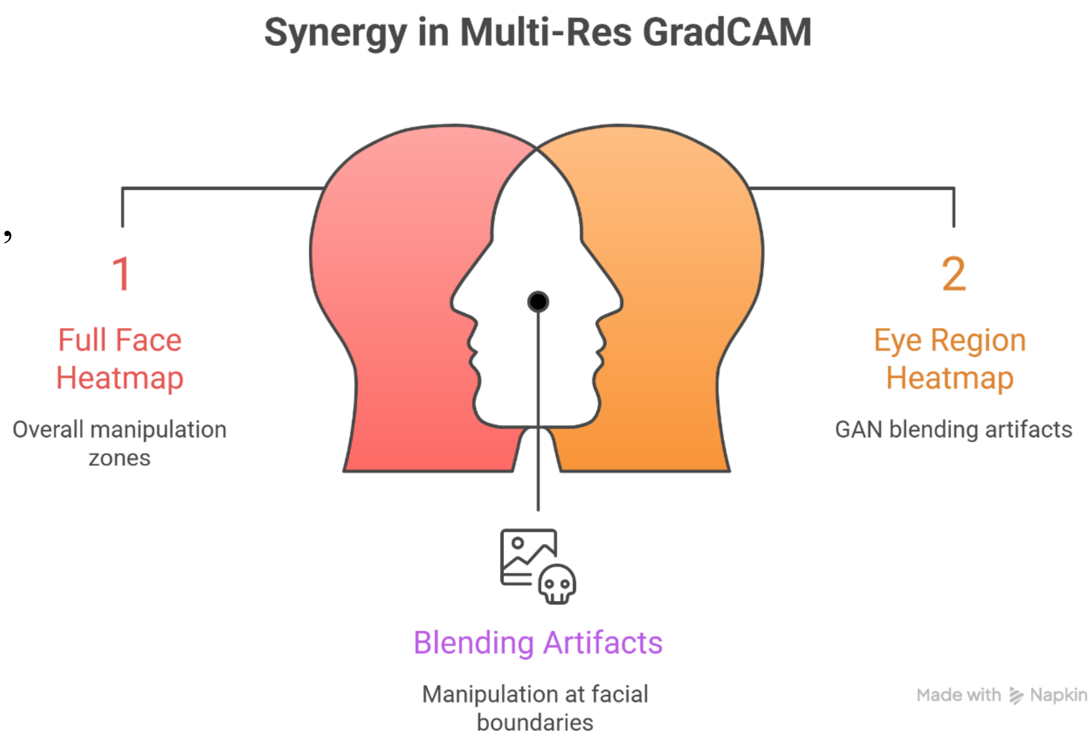
GRAD-CAM

Grad-CAM (Gradient-weighted Class Activation Mapping) is a visualization technique used to understand how CNN models make decisions.

- It highlights the important regions in an image that influence the model's prediction.
- It works by computing gradients of the target class with respect to the last convolutional layer.
- The output is a heatmap where red/yellow shows high importance and blue shows low importance.
- In deepfake detection, it helps identify manipulated facial regions like eyes, lips, and skin texture.

Why It Matters for Deepfakes

- **GAN/Diffusion fakes leave artifacts** at specific facial regions — eye boundaries, mouth blending seams, skin texture edges
- **Without GradCAM**, the model just says "Fake" — no evidence, no trust, not usable in court or KYC
- **With GradCAM**, you can visually prove *why* it's fake — judges, banks, and forensics teams can verify it



9. Results

- The model accurately classifies images as **real or deepfake** using ensemble learning.
- Performs well on **various deepfake types and image qualities**.
- Highlights **manipulated regions (eyes, lips, facial texture)** for better understanding.
- Provides **instant results** through a simple and user-friendly interface.
- Combining multiple models increases **confidence and reduces errors**.

10. Conclusion and Future Work

Conclusion:-

- Developed a deep learning-based system to detect **real and deepfake images**
- Achieved high accuracy using **ensemble models (EfficientNet, ResNet, Xception)**
- Integrated **Grad-CAM** to improve model explainability
- System is **reliable, efficient, and suitable for real-world applications**

Future work :-

- Improve accuracy using larger datasets and extend detection to videos and live streams.
- Enhance the system with multimodal analysis (audio + video) and integrate with social media platforms.

References

[1] F. Jia and L. Z. Kong, “Intrusion Detection Algorithm Based on Convolutional Neural Network,” "Beijing Ligong Daxue Xuebao/Transaction Beijing Inst. Technol., vol. 37, no. 12, pp. 1271–1275, 2017, doi: 10.15918/j.tbit1001-0645.2017".12.011.

[2] L. Mohammadpour, T. C. Ling, C. S. Liew, and C. Y. Chong, “A Convolutional Neural Network for Network Intrusion Detection System,” "Proc. Asia-Pacific Adv. Netw., vol. 46, no. 0, pp. 50–55", 2018.

[3] U. Gur Çekmez et al., “Derin Ö˘ grenme ile A˘ g Anomali Tespiti Network Anomaly Detection with Deep Learning Ozgur Koray Sahingoz,” 2018, pp. 1–4.

[4] A. Chawla, B. Lee, S. Fallon, and P. Jacob, “Host Based Intrusion Detection System with Combined CNN/RNN Model,” "Lect. Notes Comput. Sci. (including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics), vol. 11329 LNAI", pp. 149–158, 2019.

[5] S. Naseer and Y. Saleem, “Enhanced network intrusion detection using deep convolutional neural networks,” "KSII Trans. Internet Inf. Syst., vol. 12, no. 10, pp. 5159–5178, 2018, doi: 10.3837/tiis.2018".10.028.

Thank You

Questions